Are you ready for this? An evaluation of motor-readiness cortical potentials while co-lifting with an unreliable robot partner.

Robert J. Spenceley & Ranjana K. Mehta PhD.

NeuroErgonomics Laboratory

Introduction

- Heavy objects require multiple workers to share load and coordinate lifitng[1]
- Co-lifts require a blend of[2]
  - Explicit Cues
  - Implicit Cues
- Robots can improve ergonomic quality of lifts, but little work has been done on internal representations of robot actions[3]
- In the present work, we use EEG to evaluate:
  - Motor Anticipation: Contingent Negative Variation (CNV)[4]
  - Motor Readiness: Lateral Readiness potentials (LRP)[5]
- We hypothesize that an unreliable robot will elicit larger amplitudes of both CNV and LRP in the human partner.

Method

Reliable Trials

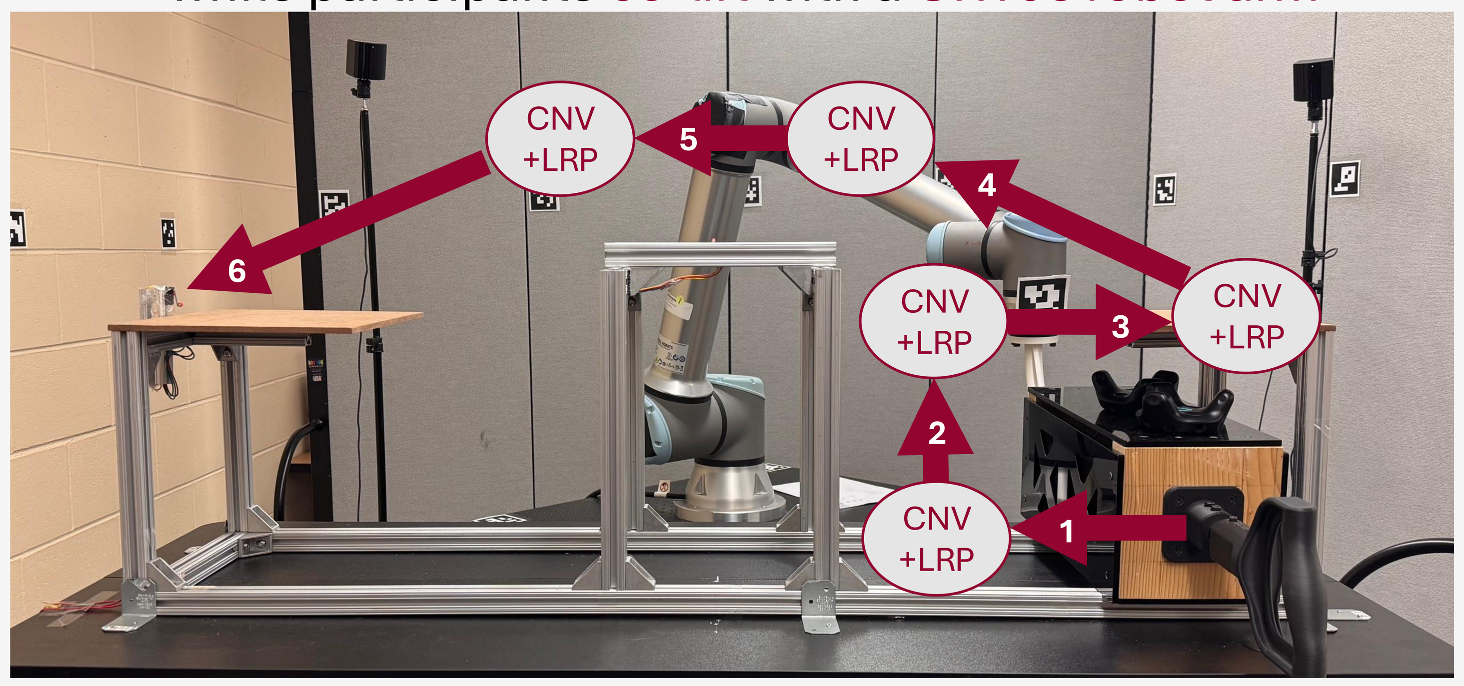
Unreliable Trials

- 40 participants (F = 20, M = 20) completed a co-lifting task with a UR10e robot arm
- 32 channel EEG cap

Subjective Measure	Anchors
"I can trust my partner during the lift"	1 (Completely Disagree) to 7 (Completely Agree)
"Rate the extent to which you and your partner coordinated effectively to achieve the desired performance"	1 (Highly Inconsistent) to 10 (Highly Consistent)
"Rate the quality of the part transport"	1 (Very rough / Jerky) to 5 (Very smooth / No Jerk)

Experimental Task

ERPs collected across reliable and unreliable conditions, while participants co-lift with a UR10e robot arm



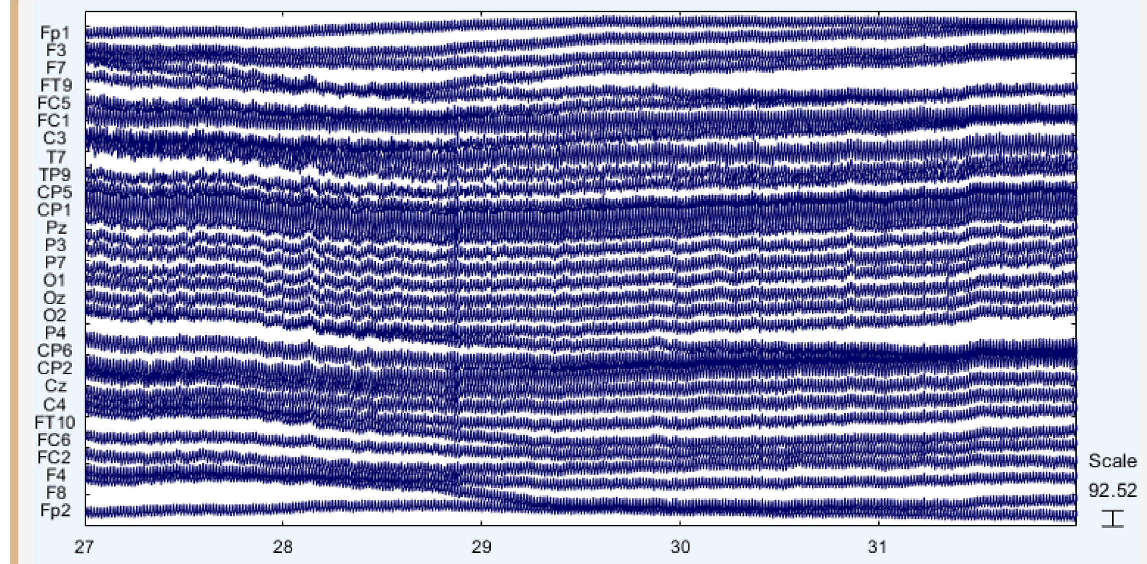
Reliable trials: fixed movement speed of 0.2 *m/s*

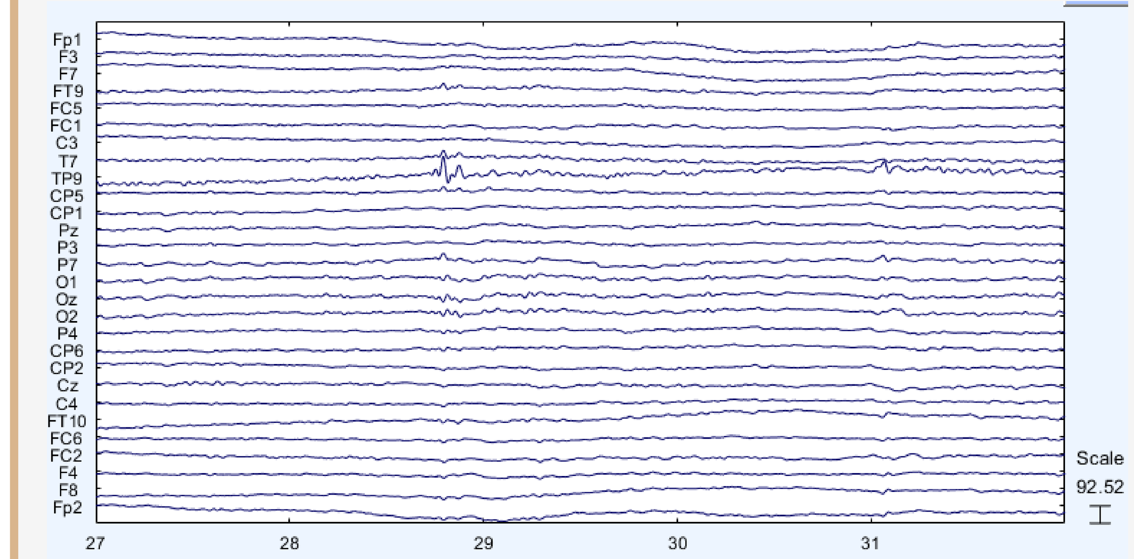
Unreliable trials: 0.2 *m/s* in segments 1, 2, & 6, 0.4 *m/s* in segments 3 & 4, and 0.1 *m/s* in segment 5

Contingent negative variation (CNV) was collected as a measure of **motor anticipation**[4] **-500 *ms* – 0 *ms* at Fz and Cz**

Lateral readiness potentials (LRP) were collected as a measure of **motor readiness**[5] **-250 *ms* – 0 *ms* at C3 – C4**

Raw EEG





Preprocessed EEG

Preprocessed Steps[6]

1. Manual rejection
2. 0.1 – 30 Hz IIR Butterworth
3. Independent component Analysis + bad channel rejection
4. Re-referenced (average)

Software

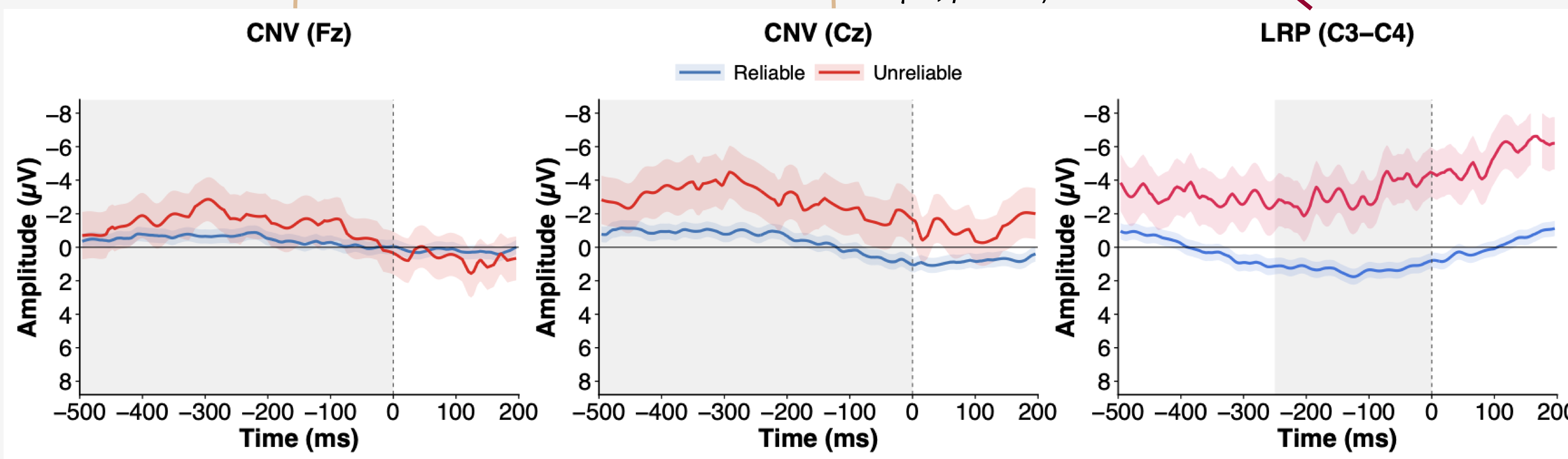
- MATLAB (v. R2024b)
- EEGLAB (v. 2024.0)
- ERPLAB (v. 10.1)
- Signal Processing Toolbox (v. 24.2)

11 Participants Excluded

Results

No significant differences were observed in CNV mean amplitude across Fz and Cz (all *p* > .23).

A significant difference was observed between LRP mean amplitude in the reliable and unreliable robot conditions (M_R = 1.18 ± 3.98 μV, M_{UR} = -2.5 ± 9.44 μV; *p* = .04).



Discussion

- Observed discrepancy between motor readiness and anticipation
  - Compensatory monitoring rather than failure prevention
- Participants do not anticipate failure, but remain ready to act
- Introduces LRP as a plausible method for real-time behavior adaptation

Limitations

- Small Sample size
- Few Unreliable Trials

Subjective Measures

Significant differences reported in:

- Trust

  M_R = 5.84 ± 1.34; M_{UR} = 4.88 ± 1.55; *p* < .001
- Team Coordination

  M_R = 8.01 ± 1.76, M_{UR} = 4.88 ± 7.40, *p* < .001
- Transport Quality

  M_R = 3.90 ± 0.62, M_{UR} = 3.69 ± 0.571, *p* = .004

References

1. Shaw, K., Salmon, J. L., & Killpack, M. D. (2025). Understanding human co-manipulation via motion and haptic information to enable future physical human-robotic collaborations. *Frontiers in neurorobotics*, *19*, 1480399. https://doi.org/10.3389/fnbot.2025.1480399
2. Sebanz, N., Bekkering, H., & Knoblich, G. (2006). Joint action: Bodies and minds moving together. *Trends in Cognitive Sciences, 10*(2), 70–76. https://doi.org/10.1016/j.tics.2005.12.009
3. Peshkin, M. A., Colgate, J. E., Wannasuphoprasit, W., Moore, C. A., Gillespie, R. B., & Akella, P. (2001). Cobot architecture. *IEEE Transactions on Robotics and Automation, 17*(4), 377–390. https://doi.org/10.1109/70.954751
4. Walter, W. G., Cooper, R., Aldridge, V. J., McCallum, W. C., & Winter, A. L. (1964). Contingent negative variation: an electric sign of sensori-motor association and expectancy in the human brain. *nature*, *203*(4943), 380-384.
5. van Vugt, M. K., Simen, P., Nystrom, L., Holmes, P., & Cohen, J. D. (2014). Lateralized readiness potentials reveal properties of a neural mechanism for implementing a decision threshold. *PloS one*, *9*(3), e90943. https://doi.org/10.1371/journal.pone.0090943
6. Luck, S. J. (2014). An Introduction to the Event-Related Potential Technique, 2nd Edn. Cambridge,MA:MITPress.